EMMANUEL TIGOUE, CISSP

AI Security Engineer | Security Engineering & Agentic Systems

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CERTIFICATIONS

CISSP | **SecurityX** | **CCNA** | **Security+** | AWS Security Specialty (in progress)

EXPERIENCE

AI Security Engineer | CoreDirective

Atlanta, GA | Sep 2025 - Present

- Built Knight, an **autonomous red team agent** designed to chain tools from recon to exploitation, and shipped its policy guardrails first: signed scopes, autonomy levels, and signed exploit approvals.
- Architected Squire, an AI SOC analyst that drafts a triage verdict for every alert and cuts manual triage time per alert **80%**. Runs on LangGraph with human approval gates, pgvector RAG, NeMo Guardrails PII redaction, and Langfuse tracing.
- Reviewed AI security architecture for organizations and threat modeled the AI gateway to Claude Mythos and its MCP tool access against the OWASP LLM Top 10 and MITRE ATLAS, setting the **guardrails** it shipped with.
- Ran red team assessments on the platform's production AI gateway and web applications: prompt injection, jailbreaks, data leakage, excessive agency, and OWASP web risks. Remediated all **6 high and medium findings** in **5 attack classes** before launch.
- Embedded Semgrep, Trivy, Gitleaks, OPA and OWASP ZAP checks into CI/CD, and signed every container image with Cosign so **unsigned images could not deploy**.
- Placed every admin surface behind Cloudflare Zero Trust, defined in Terraform with **zero public ports** and separate policies for human and service accounts.
- Tuned Falco eBPF container runtime detection shipped to Datadog and mapped to MITRE ATT&CK, cutting daily alerts from **200+ to 12** without losing coverage.
- Authored **57 GRC documents** mapping **140 NIST 800-53 controls** plus NIST AI RMF, SOC 2, and ISO 27001, and built a Promptfoo eval harness that regression tests the AI guardrail controls.

IT Security & Operations Manager | Texaco

Atlanta, GA | Mar 2022 - Mar 2026

- Deployed Splunk SIEM across **3** sites, centralizing FortiGate, Cisco switch, and Windows logs with on call alerting and a one year AWS S3 archive under Object Lock. Detection time fell from **48 hours to under 4**.
- Owned incident response for **45** endpoints, covering card skimming, credential compromise, and unauthorized vendor access. Wrote a **6** step IR runbook that reduced containment from **8 hours to 90 minutes**.
- Traced **$18K a year** in card fraud to skimming on outdated, tamper prone payment terminals and drove their replacement with current models on automatic vendor updates.
- Segmented a flat network into **6** VLANs behind FortiGate zones with default deny, isolating payment systems and tank gauges. Verified with Nmap and closed **12 of 14** critical PCI DSS gaps.
- Established a quarterly Nessus vulnerability management cycle with patch reporting automated in Python and PowerShell, and brought critical patching **from 90 to 30 days**.
- Removed standing admin rights and hardened Active Directory and Entra ID with MFA, FIDO2 admin keys, GPO baselines, stale account cleanup, and credential rotation.

TECHNICAL SKILLS

AI Security: LLM guardrail design, AI threat modeling, agentic architecture review, OWASP LLM Top 10, Model Context Protocol (MCP)

Cloud Security & Automation: AWS, Oracle Cloud, Docker, HashiCorp Vault, OPA policy gates, n8n, Ollama

Identity & Access Management (IAM): Teleport JIT PAM, Keycloak SSO, RBAC, least privilege design, mTLS

Frameworks: NIST AI RMF, NIST 800-53, ISO 42001, SOC 2, ISO 27001, PCI DSS

EDUCATION

Georgia State University, Andrew Young School of Policy Studies

B.A. in Economics

A.S. in Business Administration | Dean's List